

Method overview to be redrawn.
Body-Anchored Metric Calibration $\rightarrow$ Dynamic Interaction Grounding

Figure 2: GroundedHuman infers metric calibration and human placement from implicit interaction representations. Body-Anchored Scene Query encodes human–scene correspondence, whose metric readout calibrates scene depth and camera translation. Within this reference, Hierarchical Context Attention encodes regional relations, whose placement readout updates whole-body translation. Updating the body changes its relation to the scene, so the regional representation is recomputed before the next refinement step.

3 METHOD

Given a monocular video $\mathcal{I} = \{\mathbf{I}_t\}_{t=1}^T$, GroundedHuman recovers scene geometry, camera motion, and human meshes in a shared metric world coordinate system. We formulate human–scene grounding as learning *implicit interaction representations*, from which metric calibration and human placement are inferred. These representations encode how scene observations relate to a body’s metric extent and current spatial configuration. Human structure determines where and how to query the scene; attention integrates the relevant evidence into a relational representation, and geometric readouts convert it into corrections. The organizing principle is

$$(\mathcal{H}, \mathcal{S}) \xrightarrow{\text{Query / Attention}} \mathbf{Z}_{\text{int}} \xrightarrow{\text{Geometric readout}} (\text{Metric, Placement}),$$

where $\mathcal{H}$ and $\mathcal{S}$ denote human and scene observations. Interaction is implicit in the learned relation features $\mathbf{Z}_{\text{int}}$ and the corrections they support, without an intermediate contact-label prediction.

We instantiate this principle with two successive modules and their respective representations and readouts (Fig. 2). **Body-Anchored Metric Calibration** uses *Body-Anchored Scene Query* to encode human–scene correspondence and read out residual metric corrections. **Dynamic Interaction Grounding** uses *Hierarchical Context Attention* to encode regional human–environment relations and read out placement updates. Their ordering addresses an ambiguity: the same local depth discrepancy may reflect a scene-scale error or a misplaced human. Establishing the common metric reference first makes regional relations comparable for placement reasoning. Each placement update then changes the relations themselves, prompting the interaction representation to be recomputed.

Initial observations. Frozen VGGT- Ω provides relative Z-depth D_t^r , intrinsics $\mathbf{K}_t$, extrinsics $(\mathbf{R}_{cw,t}, \mathbf{t}_{cw,t}^r)$, and multi-layer scene features $\mathbf{F}_t$. Under the same intrinsics, frozen NLF provides metric SMPL observations: pose θ_{tq} , shape β_{tq} , root translation $\tau_{tq}^{(0)}$, and confidence c_{tq} for person q . Let $\bar{\mathbf{V}}_{tq}$ denote the posed vertices before root translation; the current camera-space body is $\mathbf{V}_{tq}^{c,(k)} = \bar{\mathbf{V}}_{tq} + \tau_{tq}^{(k)}$, with translation broadcast over vertices. These estimates supply the human metric prior; pose and shape are preserved throughout both modules.

3.1 BODY-ANCHORED METRIC CALIBRATION

The first module encodes the correspondence between metric human observations and relative scene geometry. Surface depth ratios establish a coarse reference; body-anchored queries then form a relation representation from which the remaining metric correction is predicted.

Analytic Scale Initialization. Projecting an initial body vertex into D_t^r yields a pair of depths along the same camera ray. We pool valid depth ratios across people:

$$s_t^c = \text{median}_{(q,i) \in \mathcal{A}_t} \frac{[\mathbf{V}_{tqi}^{c,(0)}]_z}{D_t^r \left(\pi(\mathbf{K}_t, \mathbf{V}_{tqi}^{c,(0)}) \right)}, \quad D_t^{\text{coarse}} = s_t^c D_t^r. \quad (1)$$

Here, π is perspective projection and $\mathcal{A}_t$ retains valid correspondences after projection and depth filtering, keeping the nearest sampled human surface when projections coincide. The median reduces sensitivity to isolated errors. Sampling and missing-observation handling are detailed in Appendix C.

Body-Anchored Scene Query. An anchor specifies both *where* to retrieve scene evidence and *how* to interpret it relative to the body. We construct 24 anchor tokens per person from the body state, anchor position and projection, and a nearby observation from D_t^{coarse} : its 3D position, normal, anchor-relative offset, distance, and depth residual. A 3×3 neighborhood of $\mathbf{F}_t$ supplies scene tokens $\mathbf{S}_{tqj}$ for anchor j . Self-attention relates evidence across the body, while cross-attention associates each anchor with its local scene context. Let $\mathbf{H}_{tq}^{(l)}$ denote the anchor tokens at layer l :

$$\tilde{\mathbf{H}}_{tq}^{(l)} = \text{SelfAttn}(\mathbf{H}_{tq}^{(l)}), \quad \mathbf{h}_{tqj}^{(l+1)} = \text{CrossAttn}(\tilde{\mathbf{h}}_{tqj}^{(l)}, \mathbf{S}_{tqj}). \quad (2)$$

In cross-attention, body anchors supply the queries and scene tokens supply the keys and values; normalization, residual connections, and feed-forward blocks are omitted for clarity. The resulting tokens encode body-conditioned scene evidence. Pooling the final responses gives the metric interaction representation $\mathbf{z}_{tq}^{\text{met}} = \text{Pool}(\mathbf{H}_{tq}^{(L)})$. A metric readout maps this representation to residual log-scale $\Delta \ell_{tq}$ and depth bias Δb_{tq} . Thus geometric correction is inferred from the learned human–scene relation, with analytic initialization accounting for the dominant scale mismatch.

Temporal Aggregation. Normalized NLF confidences combine person-level predictions into frame-level residual scale ρ_t and bias b_t (Appendix C). Assuming a common scene scale within the input clip, we aggregate

$$\begin{aligned} s^{\text{vid}} &= \exp \left[\text{median}_t \log(s_t^c \rho_t) \right], & b^{\text{vid}} &= \text{median}_t b_t, \\ D_t^m &= s^{\text{vid}} D_t^r + b^{\text{vid}}, & \mathbf{t}_{cw,t}^m &= s^{\text{vid}} \mathbf{t}_{cw,t}^r. \end{aligned} \quad (3)$$

Only depth receives the additive bias; camera rotation and intrinsics are unchanged. Back-projecting D_t^m with $\mathbf{K}_t$ gives camera-space metric points $\mathbf{P}_t^m$, the reference for subsequent placement refinement.

3.2 DYNAMIC INTERACTION GROUNDING

With the metric reference established, the second module encodes how a body is situated relative to its surroundings and reads out a translation correction. Moving the body changes the relevant observations and geometric residuals. Its interaction representation must therefore depend on the current placement and be recomputed as that placement is refined.

Hierarchical Context Attention. We partition the SMPL surface into 96 corresponding regions, each represented by its centroid and eight sampled vertices. At iteration k , each region gathers scene points from four supports, $\mathcal{W} = \{\mathcal{W}_3, \mathcal{W}_7, \mathcal{W}_{\text{adp}}, \mathcal{W}_{\text{ann}}\}$: fixed 3×3 and 7×7 windows, an adaptive window following its projected extent, and an outer annulus. These supports expose fine alignment cues together with broader environmental structure.

Each sample is encoded relative to the current body region using 3D and image-plane offsets and depth residuals. Comparison with an instance-aware human depth buffer separates observations associated with the person’s own surface from environmental evidence. Other-human samples are excluded from both attention channels and retained only as interference statistics. Masked attention selects and pools the relation features within each support and channel, yielding eight context tokens per region together with valid-support ratios. Empty supports return zero tokens. Fusing this context with region geometry, body-region embeddings, and interference statistics gives a regional interaction representation $\mathbf{z}_r^{\text{place},(k)}$. It encodes the body’s agreement with observed human surfaces and its spatial relationship to the environment, supplying complementary evidence for placement. The pooling formulation is provided in Appendix C.

Interaction-Aware Placement. A placement readout maps $\mathbf{z}_r^{\text{place},(k)}$ to a bounded translation proposal $\mathbf{v}_r^{(k)}$, gate $g_r^{(k)}$, and scalar log-variance $\ell_r^{(k)}$. This connects the implicit interaction representation to a concrete geometric action. Proposals are expressed along the region’s viewing direction

and two tangent directions, then converted to camera coordinates. Each proposes a translation of the *whole body*. With $m_r^{(k)}$ indicating valid support, we aggregate

$$\alpha_r^{(k)} = m_r^{(k)} g_r^{(k)} \exp(-\ell_r^{(k)}), \quad \hat{\mathbf{v}}^{(k)} = \frac{\sum_r \alpha_r^{(k)} \mathbf{v}_r^{(k)}}{\max(\sum_r \alpha_r^{(k)}, \varepsilon)}. \quad (4)$$

The readout favors supported, reliable evidence. Supervision on these corrections trains the interaction representation without requiring explicit contact labels.

Translation Refinement. A person-level gate modulates the aggregated proposal after component-wise clipping (Appendix C), producing the root update $\Delta\tau_{tq}^{(k)}$:

$$\tau_{tq}^{(k+1)} = \tau_{tq}^{(k)} + \Delta\tau_{tq}^{(k)}. \quad (5)$$

Two feed-forward steps share parameters. After each update, we rebuild region locations, sample ownership, and attention context to obtain a new $\mathbf{z}_r^{\text{place},(k+1)}$. This closes the loop between relation encoding and geometric readout: an update changes the human–scene relationship on which the next update depends. Here, *dynamic* refers to this placement-dependent recomputation. Scene geometry, camera parameters, body pose, and shape remain fixed. Finally, the calibrated inverse camera extrinsics map the refined bodies and metric scene points to a shared world frame (Appendix B).

3.3 TRAINING OBJECTIVES

The two stages are trained separately, with VGGT- Ω and NLF frozen. Metric calibration uses metrically supervised sequences with synthetic scale perturbations. Initial scale training is followed by training on analytically initialized states, supervising the residual correction required from that state:

$$\mathcal{L}_{\text{calib}} = \lambda_s \mathcal{L}_{\text{scale}} + \lambda_b \mathcal{L}_{\text{bias}} + \lambda_d \mathcal{L}_{\text{depth}}. \quad (6)$$

The terms supervise residual log-scale, depth bias, and calibrated depth. In the residual-head training stage, the anchor interaction backbone is frozen and only the scale and bias heads are updated. The bias target is zero for this scale-perturbation path, constraining additive compensation for multiplicative errors.

With calibration fixed, Dynamic Interaction Grounding is trained on clean, scale-perturbed, translation-perturbed, and mixed states:

$$\mathcal{L}_{\text{ground}} = \lambda_v \mathcal{L}_{\text{vote}} + \lambda_t \mathcal{L}_{\text{trans}} + \lambda_g \mathcal{L}_{\text{gate}} + \lambda_r \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{state}}. \quad (7)$$

Regional votes target the remaining displacement divided by the number of remaining updates. Final-translation supervision trains the placement readout, gate supervision trains update validity, and regularization controls update magnitude. State-dependent terms encourage recovery from large errors, preservation of accurate inputs, and non-degrading corrections across iterations. Appendix D details the objectives and training configuration status.